

Redirect, Forward & HTTP Status Codes — Assignment Answers

Topics: `sendRedirect()`, `RequestDispatcher.forward()`, `setStatus()`, 404/302/200

Q1. MovieLoginServlet — Login + `sendRedirect()` to Dashboard

```
import java.io.IOException;
import jakarta.servlet.ServletException;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;

public class MovieLoginServlet extends HttpServlet {

    // Hardcoded credentials for this demo
    private static final String VALID_USERNAME = "admin";
    private static final String VALID_PASSWORD = "movie123";

    protected void doPost(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

        String username = request.getParameter("username");
        String password = request.getParameter("password");

        if (VALID_USERNAME.equals(username) && VALID_PASSWORD.equals(password)) {
            // Successful login -> redirect browser to a new servlet/URL
            response.sendRedirect("MovieDashboardServlet");
        } else {
            response.setContentType("text/html");
            response.getWriter().println("<h3>Invalid username or password</h3>");
        }
    }
}
```

MovieDashboardServlet simply displays a welcome/dashboard message when the browser lands on it after the redirect:

```
public class MovieDashboardServlet extends HttpServlet {
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
        response.setContentType("text/html");
        response.getWriter().println("<h2>Welcome to your Movie Dashboard!</h2>");
    }
}
```

Note: `sendRedirect()` sends an HTTP 302 response telling the browser to make a brand-new request to MovieDashboardServlet — the URL in the address bar changes.

Q2. PlaylistServlet → PlaylistDetailsServlet via RequestDispatcher.forward()

```
import java.io.IOException;
import jakarta.servlet.RequestDispatcher;
import jakarta.servlet.ServletException;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;

public class PlaylistServlet extends HttpServlet {

    protected void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

        String playlistId = request.getParameter("playlistId");

        // Forward the SAME request/response objects to another servlet
        RequestDispatcher dispatcher = request.getRequestDispatcher("PlaylistDetailsServlet");
        dispatcher.forward(request, response);
    }
}
```

PlaylistDetailsServlet.java:

```
public class PlaylistDetailsServlet extends HttpServlet {

    protected void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

        String playlistId = request.getParameter("playlistId");

        response.setContentType("text/html");
        response.getWriter().println("<h2>Playlist Details for ID: " + playlistId + "</h2>");
    }
}
```

Note: forward() happens entirely on the server — the browser's URL stays as PlaylistServlet, and the original request parameters (like playlistId) remain available in the target servlet.

Q3. ProductServlet — 404 / 200 Status Codes

```
import java.io.IOException;
import jakarta.servlet.ServletException;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;

public class ProductServlet extends HttpServlet {

    protected void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

        String productId = request.getParameter("productId");
        response.setContentType("text/html");

        boolean isValid = (productId != null) && !productId.trim().isEmpty()
            && productId.matches("[A-Za-z0-9]+");

        if (!isValid) {
            response.setStatus(HttpServletResponse.SC_NOT_FOUND); // 404
            response.getWriter().println("<h2>Product not found</h2>");
        } else {
            response.setStatus(HttpServletResponse.SC_OK); // 200
            response.getWriter().println("<h2>Product details for " + productId + "</h2>");
        }
    }
}
```

Example: /ProductServlet?productId=SHOE123 → status 200, 'Product details for SHOE123'.
/ProductServlet (no productId) or an invalid value → status 404, 'Product not found'.

Q4. InstaRedirectServlet — setStatus(302) + Logging + sendRedirect()

```
import java.io.IOException;
import jakarta.servlet.ServletException;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;

public class InstaRedirectServlet extends HttpServlet {

    protected void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

        response.setStatus(HttpServletResponse.SC_FOUND); // sets status to 302
        System.out.println("Redirecting user to TrendingServlet (HTTP 302)");

        response.sendRedirect("TrendingServlet");
    }
}
```

Note: `sendRedirect()` already sets the status to 302 internally, so the explicit `setStatus(302)` call here is mainly for clarity/logging purposes in this exercise — calling it before `sendRedirect()` does not change the outcome, since `sendRedirect()` will set/override the status and Location header itself.

<code>sendRedirect()</code>	<code>RequestDispatcher.forward()</code>
New request from the browser (client-side)	Same request, handled server-side
URL in address bar changes	URL stays the same
Sends HTTP 302 to the browser	No extra HTTP round-trip
Can redirect to any URL, even external sites	Only within the same web application